

NLP For Vis: Designing a Chart to Alt Text Model

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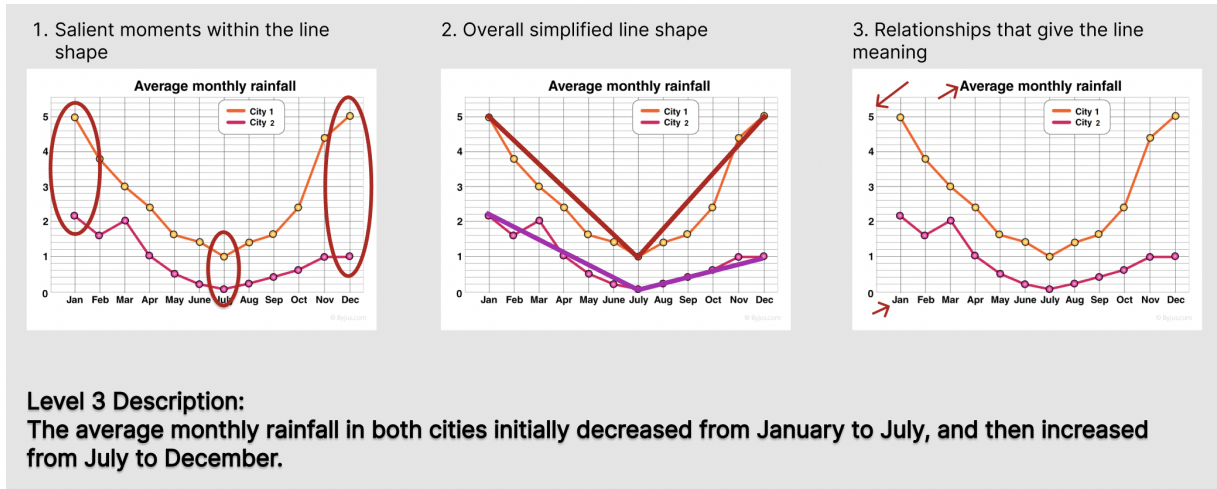


Figure 1: 3-stage NLP model that produces a level 3 textual description

ABSTRACT

We present a design contribution for an NLP model capable of producing a textual description of a data visualization which captures higher level trends in the data. Our design process breaks model heuristics into three stages based on visual salient features and the relationships between geometric and textual elements. While our model pushes the state of the art (SotA) in automated chart descriptions, we have pessimistic considerations for the ethical and functional implications of its use. Our core hypothesis is: Can we assume this SotA model will still fail to adequately meet the needs of people with disabilities? And then, can this assumption help us de-risk this work? Are there advantages to a pessimistic approach in regards to model design and ethical considerations?

1 INTRODUCTION

Addressing the inaccessibility of data visualization is growing in prominence and importance in our field of work. Some attempts have been made to improve resources for practitioners [3] or build more accessible visualization tooling or techniques [8]. Despite these attempts, fundamental issues remain for addressing the growing scale of inaccessibility. To address scale, the most common technique is to use NLP and machine learning to automatically describe charts or produce analytical insights [2]. While this growing body of automated description work holds promise, there are significant ethical and access considerations that remain when describing charts and graphs [1, 5] as well as potential interpretive issues [7].

Pushing the state of the art in machine learning when accessibility is at stake should be held to a higher standard of scrutiny in evaluation. In particular, many existing state of the art models focus on

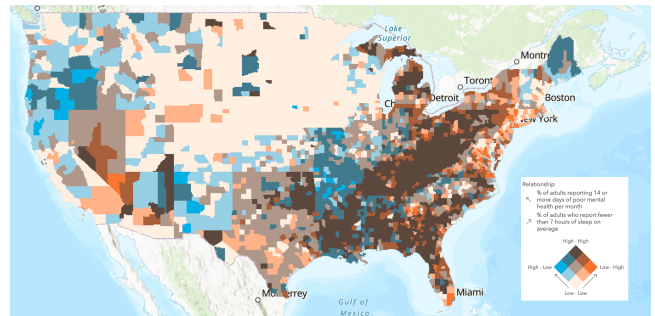


Figure 2: Bivariate map used in data collection

providing only surface level descriptions of charts, rather than detailing important trends in the data. Our project is building towards an evaluation using replay enactments, which is a way to test machine models in combination with a wizard-of-oz approach [4]. We set out to design a machine learning model that pushes the state of the art, capable of producing a level 3 semantic description, which aims to describe trends and summaries across sub-categories in the data [6]. But we are also working towards an evaluation of this model using blind and visually impaired participants in a design workshop. We intend to both:

1. Improve the current state of the art in automated chart descriptions.
2. Remain pessimistic about our model's ability to provide meaningful access for people with disabilities.

2 GATHERING DATA AND TESTING THE FIELD

We first gathered ten charts of various types and subjects such as the one in Figure 2, primarily from the DataIsBeautiful reddit.

These included line charts, bar charts, pie charts, and even a bivariate map (Figure 2). We then looked for existing models that

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would translate a visual chart to a text description, and tested them on these charts. There were two models we used for this purpose: Microsoft Word’s built-in text description generator, and the Seeing AI app. In Microsoft Word, the description generator would automatically generate a text description when a chart was inserted. In the Seeing AI app, there was a camera that we would point at the screen with the chart, and from there, it would automatically generate a description based on the chart it processed. A couple things to note about the Seeing AI app are that it was not designed specifically for charts, and was also designed by Microsoft. Overall, we found that both models were very unsatisfactory in their descriptions of the charts. In all of the descriptions, both models focused primarily on the type of chart that they processed, which were even sometimes inaccurate. For example, for the bivariate chart, the Seeing AI app described it as “a map of the world” even though the chart only included the United States. None of the descriptions mentioned anything about the actual data in any of the charts and most of the time, were not very specific about the type of chart. These descriptions were mostly along the lines of “diagram” or “chart,” occasionally being a bit more specific such as “pie chart” or “line chart.”

3 NARROWING THE SCOPE

From the ten charts that we had originally gathered, we narrowed our scope to focus only on line charts. There were a few reasons for this. Firstly, line charts are extremely common and are not too complicated. As a result, focusing on these specific types of charts would not only make our work more manageable, but it would also still be very impactful for most people. Line charts also have a higher level of visual presentation, meaning there is usually a trend that we want to describe. In addition, multi-series lines, which constituted most of our charts, builds more expectations for level 3 descriptions, as there is not only a trend within one line, but also trends across the multiple lines. Consequently, there is more to unpack with these charts that could also lead to a potential point of failure for many chart-to-text models.

4 MODEL DESIGN

We then tried to annotate parts of the line chart in different stages using Figma, focusing on what information we might use to produce a Level 3 description. We first identified salient features along each line in the chart, specifically the minimum, maximum, starting, and ending points of the lines. We then annotated the overall simplified line shape that retains those salient features in the second step. The third step was annotating parts of the chart that added meaning to the lines, such as the legend, axis values and labels (which were not always present), and the title. Our model then tries to understand the trend of each line based on the annotated salient features, overall line shape, and the annotated areas that add meaning. It will then try to understand commonalities between each of the trends, for example if they all go down and then up or if most of them share a trend, or if there is no noticeable trend at all. These steps are illustrated in Figure 1.

5 RESULTS

We found that our space is much more complicated than we originally thought, and there are multiple points of failure that are important to discuss in order to effectively build a good model. One primary issue to address is whether the model can identify different shapes and then know to look for trends vs clusters along a single shape. For example, in Figure 3, there aren’t any meaningful relationships between the lines, but rather each line itself has its own separate meaning. In this case, the model must be able to identify whether to focus on trends between the lines or each line’s individual trend.

Another point of discussion is whether the model can identify text from the title, legend, and axes and connect them to shapes

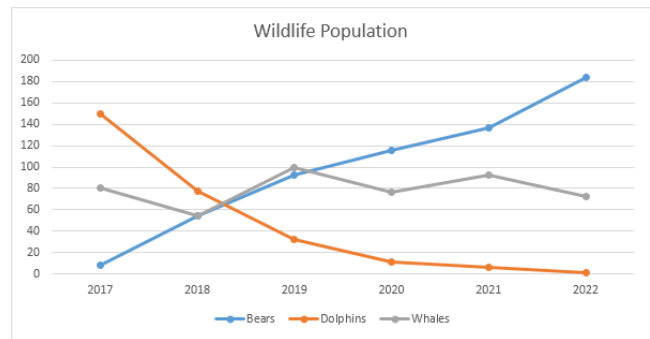


Figure 3: Another example line chart used as input for our model

in order to provide a more meaningful description. We then want to identify whether the model can then use the text semantics and describe higher level trends.

6 FUTURE WORK

Now that we have created a model design and identified certain points of failure for the model, the next steps would be to actually build a model that incorporates some of these ideas. We would then want to run user sessions to evaluate the effectiveness of such a model, using both good and bad descriptions from the model in order to measure the ethical impact of such models through replay enactments.

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